EXP: 12(a) ECHO CLIENT-SERVER USING TCP.

24-09-24

AIM:-

To implement an Echo Client-Server application using TCP sockets and verify communication between the client and server.

PROCEDURE:-

1- Create a TCP Echo Server Using Server Socket and socket.

2- Assign a port number to the server.

3. Compile and execute the TCP Server program.

4. Create a TCP Echo Client using socket.

5. Connect the client to the server using the server address and port number.

6. Enter a message through the passage and send it back to the client.

7. The server recieves the passage and send it back to the client.

8. Display the echoed message at the client.

9. Enter bye to terminate the communication.

RESULT:-

Thus, the TCP Echo communication was successfully implemented and the client successfully recieved the echoed message from the server.

```
Problems  Output  Debug Console  Terminal  Ports  Jupyter

PS C:\Users\Luqman\Downloads\CN_EXPT_12> javac TCPEchoServer.java TCPEchoClient.java
PS C:\Users\Luqman\Downloads\CN_EXPT_12> java TCPEchoServer
TCP Echo Server listening on port 5000
Client connected: /127.0.0.1:50613
Received: hello
Received: I am studying in REC
```

```
Problems  Output  Debug Console  Terminal  Ports  Jupyter

PS C:\Users\Luqman\Downloads\CN_EXPT_12> java TCPEchoClient
Connected to TCP Echo Server at localhost:5000
Type messages to send (type 'bye' to quit):
hello
Server: Echo: hello
I am studying in REC
Server: Echo: I am studying in REC
```

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TCPEchoServer

```
import java.io;
import java.net;
public class ChatServer{
    public static void;
    s = socket.socket()
    s.bind("local host", 5000)
    s.listen(n)
    c, addr = s.accept()
    msg = c.recv(1024).
    c.send(msg)
    c.close()
    s.close().
}
```

TCPEchoClient.

```
import socket.
s = socket.socket
s.connect(("localhost", 8000))
s.send(b"Hello Server")
print(s.recv(1024).decode())
s.close()
```

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